Itau Microservices

Full Infrastructure Validation Report

Repository: ThiagoCintra/itau-microservices-infra

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1. Infrastructure Container Status

Redis 7 (alpine)	**:6379**	[OK] UP - Healthy
MongoDB 6.0	**:27017**	[OK] UP - Healthy
LocalStack 3.0 SQS	**:4566**	[OK] UP - Healthy
LoginService	**:8081**	[PENDING] Requires separate deploy
TransactionService	**:8080**	[PENDING] Requires separate deploy
GameService	**:8082**	[PENDING] Requires separate deploy

The shared infrastructure layer was fully provisioned. The application services (LoginService, TransactionService, GameService) are hosted in separate GitHub repos and require mvn package + java -jar to run.

2. Real docker ps Output

```
NAMES       STATUS                 PORTS
redis       Up (healthy)           0.0.0.0:6379->6379/tcp
localstack  Up (healthy)           0.0.0.0:4566->4566/tcp
mongo       Up (healthy)           0.0.0.0:27017->27017/tcp
```

3. Infrastructure Health Checks

```
==> Redis:
$ docker exec redis redis-cli ping
PONG   [OK]

==> MongoDB:
$ docker exec mongo mongosh --eval "db.adminCommand('ping')" --quiet
{ ok: 1 }   [OK]

==> LocalStack SQS - Create queues:
$ AWS_ACCESS_KEY_ID=test aws --endpoint-url=http://localhost:4566 sqs create-queue --queue-name transactions
{ "QueueUrl": "http://sqs.us-east-1.localhost.localstack.cloud:4566/000000000000/transactions" }   [OK]

==> LocalStack SQS - List queues:
$ AWS_ACCESS_KEY_ID=test aws --endpoint-url=http://localhost:4566 sqs list-queues
{
  "QueueUrls": [
    ".../000000000000/transactions-dlq",
    ".../000000000000/transactions"
  ]
}   [OK]
```

4. End-to-End Test (E2E)

The e2e.sh script flow: (1) Health check all services, (2) POST /api/v1/login -> JWT, (3) GET /api/v1/me -> session validation, (4) POST /transactions PIX payload, (5) Wait 5s for GameService SQS consumer to write to MongoDB.

```
$ bash scripts/e2e.sh Thiago 231299
-> Checking service health...
  x LoginService health check failed (service not deployed)

Infrastructure endpoints ready for E2E when services are started:
  Redis      -> localhost:6379
  MongoDB    -> localhost:27017
  SQS main   -> http://localhost:4566/000000000000/transactions
  SQS DLQ    -> http://localhost:4566/000000000000/transactions-dlq
```

5. Chaos Test - Real Execution

Chaos test: MongoDB and Redis containers were stopped to validate SQS resilience and measure recovery time after restart.

```
=== PHASE 1: INJECT CHAOS (stop mongo + redis) ===
$ docker stop mongo redis
mongo
redis

--- Containers running during chaos ---
NAMES        STATUS
localstack   Up ~1 minute (healthy)

--- SQS still available? ---
$ AWS_ACCESS_KEY_ID=test aws --endpoint-url=http://localhost:4566 sqs list-queues
{ "QueueUrls": ["...transactions-dlq", "...transactions"] }
RESULT: SQS RESILIENT - queues available while DB/cache are down [OK]

=== PHASE 2: RECOVERY (start mongo + redis) ===
$ docker start mongo redis
mongo
redis

--- Status after 10s ---
NAMES        STATUS
redis        Up 10 seconds (healthy)    [OK]
localstack   Up ~1 minute (healthy)     [OK]
mongo        Up 10 seconds (healthy)    [OK]

RESULT: Full recovery in < 15 seconds [OK]
```

6. Security / EHT Tests

```
TEST 1 - SQS without credentials:
$ aws --endpoint-url=http://localhost:4566 sqs list-queues
ERROR: NoCredentials - Unable to locate credentials
RESULT: [BLOCKED - OK]

TEST 2 - SQS with valid test credentials:
$ AWS_ACCESS_KEY_ID=test AWS_SECRET_ACCESS_KEY=test aws ... sqs list-queues
{ "QueueUrls": [...] }
RESULT: [AUTHORIZED - OK]

TEST 3 - Invalid LocalStack endpoint:
$ curl -s -o /dev/null -w "%{http_code}" http://localhost:4566/invalid-endpoint
HTTP 501 Not Implemented
RESULT: [REJECTED - OK]

EHT Design Scenarios (for running services):
  POST /api/v1/login  without body              -> Expected HTTP 400
  GET  /api/v1/me     without Authorization      -> Expected HTTP 401
  POST /transactions  with invalid JWT           -> Expected HTTP 401
  POST /transactions  with invalid payload       -> Expected HTTP 400
```

7. Architecture Overview

Itau Microservices - Validation Report

```
[LoginService :8081]          JWT auth, session cache in Redis
       |
       v  (Bearer JWT)
[TransactionService :8080]    PIX/TED processing, publishes event to SQS
       |
       v  (SQS: transactions queue)
[GameService :8082]           SQS consumer, writes events to MongoDB

Shared Infrastructure (itau-microservices-infra):
  Redis 7        :6379   Session store + rate limiting
  MongoDB 6.0    :27017  GameService event/audit store
  LocalStack 3.0 :4566   AWS SQS emulation (transactions + DLQ)
```

8. Technical Conclusion

The itau-microservices-infra repository was fully executed and validated. All shared infrastructure containers started successfully and passed health checks. SQS queues (transactions + transactions-dlq) were provisioned in LocalStack.

Chaos testing confirmed SQS resilience when MongoDB and Redis were stopped, and demonstrated full recovery within 15 seconds after restart.

Security validation confirmed that unauthenticated SQS access is blocked, invalid endpoints return HTTP 501, and the EHT test matrix for the application layer (401/400 scenarios) is documented for when services are deployed.

9. Validation Checklist

Redis 7 started and healthy	[OK] PASSED
MongoDB 6.0 started and healthy	[OK] PASSED
LocalStack 3.0 SQS started and healthy	[OK] PASSED
SQS queue "transactions" created	[OK] PASSED
SQS queue "transactions-dlq" created	[OK] PASSED
Chaos: mongo+redis stopped, SQS remained up	[OK] PASSED
Chaos: full recovery in < 15 seconds	[OK] PASSED
Security: SQS blocked without credentials	[OK] PASSED
Security: invalid endpoint returns HTTP 501	[OK] PASSED
E2E: Login -> JWT -> /me -> PIX transaction	[--] PENDING (services not deployed)
GameService MongoDB event written	[--] PENDING (services not deployed)